TDE (Tower Defence Engine.)

Introduction

Tower defence is a subgenre of real-time strategy games in which the player's main task is to protect a strategic asset (base, castle etc) from waves of advancing enemies by building defensive towers/structures along their path.
Our team is developing an engine for more comfortable development of games in the TD genre.

Concept

Our engine will provide following facilities for developing tower defence games:

- Level editor
- 2d graphics support
- Runtime library (code that actually runs the game)
- Ready-to-use base components (e.g. abstract tower, enemy, spawn point, enemy path, players' strategic asset)

The engine will consist of two parts:

1. Level editor for customization of:
    - Textures (base textures and loader for custom ones): Towers, enemies, strategic assets, game GUI, background, projectiles
    - Game logic
        - Enemies
            - Behaviour: Target logic, following the path, Damage customization (to strategic asset, to units), HP bar, Spawn logic
        - Towers
            - Tower type, Target logic, Projectiles collision, Attack range, Attack cooldown, Damage customization, Upgrade/sell system, HP bar, Active/inactive mode
    - Tower/enemies spots, enemy paths
2. Runtime library
    - Rendering
    - Game objects and execution management

These two parts will communicate in following way:

1. Game editor generates level description in some format, recognisable by the runtime
2. Also, game editor uses game logic implementation language toolchain to compile and link an executable, which will call runtime library entry point
3. The runtime will be distributed as a dynamic library, linked to the game during startup

The editor will be built around ECS (Entity Component System) architectural pattern:

1. There will be predefined set of components and systems with base functionality for developers to use
2. Developer will be able to define custom components and systems to implement some game-specific logic
3. To construct game levels developer will create entities using these components and choose which systems will be ran

Potential Users - developers (level designers and gameplay programmers(?))

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